The total possible points for HW7 are 74:

P1 (17): (a) 5 points including 3 points for answering low ethane conversion and 2 points for explanation; (b) 4 points; (c) 6 points; (d) 2 points.

P2 (15): fractional conversion of glycerine worth 5 points; "not operating in equilibrium" worth 2 points; each of the suggestion worth 2 points; mole fraction of H2 and fractional conversion of glycerine at equilibrium worth 4 points.

P3 (16). Each atom economy for those two methods worth 4 points. Calculation for two process worth 5 points. A reasonable conclusion worth 3 points. Again, you can choose either the original process or Connie method as long as you can defend your points. For example, for original method, higher atom economy is an advantage and for Connie method, elimination of erosive acetic acid is an advantage.

P4 (26). Each of the reaction worth 2 points. Proving the laboratory analysis reasonable worth 4 points. Each answer of values (a-e) worth 2 points. At equilibrium, each of the calculation for maximize (a-d) worth 2 points. I want to mention again here the advantage of using a computational tool to solve this optimization problem, though you can solve the process by plugging different values into the equations.